

Number, units and estimation - Set 1

TMUA practice questions by topic. Specification M1.1, M1.2, M2.1-M2.14. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Find the highest common factor of $2^3 \times 3^2 \times 5$ and $2^2 \times 3^3 \times 7$.

A 18
B 6
C 12
D 36
E 72
F 108

- 2 Express the recurring decimal $0.\dot{2}\dot{7}$ as a fraction in its lowest terms.

A $\frac{9}{33}$
B $\frac{18}{33}$
C $\frac{27}{11}$
D $\frac{99}{11}$
E $\frac{107}{100}$
F $\frac{27}{100}$

- 3 A length is given as 4.6 cm correct to 1 decimal place, and a width as 3 cm correct to the nearest cm. Find the upper bound of the area.

A 16.1 cm^2
B 11.375 cm^2
C 16.45 cm^2
D 16.275 cm^2
E 14.375 cm^2
F 13.8 cm^2

- 4 Evaluate $\frac{(3 \times 10^4) \times (8 \times 10^{-2})}{6 \times 10^{-3}}$, giving your answer in standard form.
- A** 4×10^4
B 4×10^6
C 4×10^{-1}
D 2.4×10^5
E 1.5×10^5
F 4×10^5
- 5 A code consists of two letters from A, B, C, D, E followed by two digits from 1 to 9. Letters may be repeated but the two digits must be different. How many codes are possible?
- A** 3240
B 2025
C 1440
D 900
E 1620
F 1800
- 6 A car travels at 90 km/h. What is this speed in metres per second?
- A** 30 m/s
B 54 m/s
C 15 m/s
D 25 m/s
E 324 m/s
F 1.5 m/s
- 7 Estimate the value of $\frac{4.92 \times \sqrt{99.3}}{0.198}$ by rounding each number to 1 significant figure.
- A** 2500
B 250
C 500
D 100
E 50
F 25

- 8** A number x is truncated (not rounded) to 2 decimal places and the result is 3.14. Which inequality describes the possible values of x ?
- A** $3.14 < x < 3.15$
- B** $3.135 \leq x < 3.145$
- C** $3.13 < x \leq 3.14$
- D** $3.135 < x < 3.145$
- E** $3.14 \leq x < 3.15$
- F** $3.14 \leq x \leq 3.15$
- 9** Evaluate $\left(2\frac{1}{3} - 1\frac{3}{4}\right) \div \frac{7}{6}$.
- A** $\frac{2}{37}$
- B** $\frac{13}{72}$
- C** $\frac{13}{72}$
- D** $\frac{2}{7}$
- E** $\frac{24}{24}$
- F** 1
- 10** Which of the following is equal to $\frac{8^4 \times 2^{-5}}{4^3}$?
- A** 1
- B** 2
- C** 16
- D** 8
- E** 4
- F** $\frac{1}{2}$
- 11** Simplify $\sqrt{48} \times \sqrt{27} \div \sqrt{12}$.
- A** 18
- B** 12
- C** $9\sqrt{3}$
- D** 6
- E** 36
- F** $6\sqrt{3}$

12 How many three-digit positive integers have digits that are all different and all odd?

A 120

B 24

C 125

D 100

E 30

F 60